

Embracing Lag Uncertainty: Bayesian Model Averaging in Set-Identified Structural VARs

Akın Aydın* Ferdinand Loser†

June 21, 2026

Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

Keywords: Structural Vector Autoregression (SVAR), Set Identification, Bayesian VAR (BVAR), Lag Order Selection, Monte Carlo Simulation

JEL Codes: C11, C15, C32, C52

*Ludwig-Maximilian University of Munich, akin.aydin@campus.lmu.de

†Ludwig-Maximilian University of Munich, ferdinand.loser@campus.lmu.de

1 Introduction

Structural Vector Autoregressions (VAR) remain the cornerstone of a big section of empirical macroeconomics, especially for monetary policy, shock transmission, and financial markets. While forecasting may be the main goal for some actors, especially for central banks or national economic institutes, many researchers use it in order to estimate the relationship between different factors in determining macroeconomic movements. For this purpose, Impulse Response Functions (IRF) are deduced from the VAR parameters, establishing a link to how a shock in one variable affects the development of another period, both contemporaneously as well as over a horizon of several months or even years. For example, it may be applied to study how shocks in speculative oil demand affect oil demand and supply flows (Kilian and Murphy, 2014) or how interest rate shocks impact credit costs (Gertler and Karadi, 2015) (change to add newer papers). The specification of the VAR model, as well as the method to decompose it into structural form, are the two crucial specifications that a researcher has to decide on. Traditionally, VAR models were specified with a lag order determined by economic theory or an information criterion (IC) and then estimated by Ordinary Least Squares (OLS) or Maximum Likelihood (ML). The past 15 years saw a surge in popularity of Bayesian methods, in which not one single lag order is specified, but lag orders are weighed and/or shrunk based on a prior as well as a posterior probability distribution. These methods enable the researcher to not only rely on one specification but to incorporate a larger model or set of models while still keeping variance low. The deduction methodology of IRFs has also developed a lot ever since ?? showed that structural IRFs may be retrieved by Cholesky decomposition, i.e., pre-determined causal ordering of the relationships between variables. Newer methods typically require weaker assumptions and follow a more agnostic approach to the inference of economic relationships. The most dominant approach nowadays is set identification using sign restrictions based on economic theory. Unlike with Cholesky decomposition, all variables can simultaneously react to each other's shocks, and sign restrictions can reduce the possible set of IRFs to only include those that fit the researcher's assumptions. In the end, there is a set of admissible IRFs of which the median target, the IRF overall closest to the median of all IRFs, is usually selected as the main result.

This study takes inspiration from Ivanov and Kilian (2005), the first paper to study which specification of a VAR model is the best at retrieving the true IRF by using the Monte Carlo simulation method. In 2005, this meant testing criteria for lag selection for how well they can fit VAR models to retrieve structural IRFs by Cholesky decomposition. We aim to update this to the more recent world of VAR methodology where IRFs are determined by set identification and estimation doesn't only rely on information criteria, but also on Bayesian methods. We construct a realistic DGP based on Kilian and Murphy (2014)'s study of the oil market, using the set identification method. Then we compare

different methods over repeated simulations in how well they can retrieve the true IRF by means of the median target of their identified set. This study focuses on comparing the information criteria AIC, BIC, and HQC with BVAR estimations using different overall shrinkage parameters λ_1 . An accompanying paper by (CITATION) also uses the popular method of Bayesian Model Averaging. We find that BVARs can significantly outperform OLS estimated VAR models; however, with strong differences in performance depending on λ_1 . Optimization of λ_1 using marginal density distribution (MDD), which optimizes on out-of-sample performance, is found not to always be able to select the best λ_1 for the purpose of IRF decomposition. We therefore show that the goal of forecasting and IRF analysis do not necessarily work together in VAR estimation.

The rest of the paper is structured as follows:

2 Simulation Design

This Monte Carlo simulation study follows Ivanov and Kilian (2005) design with several alterations in order to make our extensions feasible and to fit the needs of our analysis. First of all, we only use one monthly DGP based on the empirical VAR study of Kilian and Murphy (2014). Therefore, we do not get estimations for quarterly VAR series or VEC series, and additionally, we do not have additional robustness by including several different series with different macroeconomic indicators and movements. The main reason for this is our tight time constraint. However, this concern is also not to be overstated as Ivanov and Kilian (2005) did not find big differences between different DGPs of the same type (monthly, quarterly, VEC). We use the dataset from Kilian and Murphy (2014) containing the following 4 variables: percent change in global oil production, real activity index from Kilian (2009), the log real price of oil, and changes in OECD crude oil inventories. Just as in the original study, we first estimate a VAR with a set lag order of these 4 variables and then use the sign restrictions approach explained earlier to identify structural IRFs. Therefore, our estimation becomes much more complex and updates the findings of Ivanov and Kilian (2005) to a more modern and widely used approach in the literature. All papers used for the DGP generation in Ivanov and Kilian (2005) are from the 1990s and early 2000s and are based on the Cholesky decomposition approach for identification. Cholesky decomposition only gives one unique set of IRFs, whereas the sign restrictions approach principally has an infinite number of possible sets of IRFs.

We follow Kilian and Murphy (2014) in searching for a large number of admissible IRFs by randomly drawing from the space of rotation matrices. We use exactly the same sign restriction matrix, but limit the number of periods for which dynamic sign restrictions are imposed to 3 instead of 12, and do not include additional elasticity bounds. This is a significant diversion from the original study as it makes the identification more agnostic and opens up the space of admissible IRFs. We are therefore closer to the original sign

restrictions approach of Uhlig (2005). We understand that the additional restrictions of Kilian and Murphy (2014) are motivated by limiting the set to fit theory and other empirical evidence, we have to take the less restrictive approach in order to make the estimation feasible. Especially for low lag orders and smaller samples, these restrictions have proven to be so strong that even several million draws of the rotation matrix do not give us enough admissible IRFs to make a good inference. For the simulation, this exceedingly high number of draws would need to then be repeated hundreds of times.

Out of this set, we select the median target IRF as our true IRF, which is defined as the IRF that has the smallest distance to the median of all admissible IRFs. The DGP follows this IRF as it is generated by a structural VAR model using the structural impact matrix $\tilde{B}$ that corresponds to the median target IRF as the parameter for the structural shocks e_t , which is drawn from an i.i.d. standard normal distribution. The formula for the DGP is represented by a structural VAR model using the estimated reduced form parameters A (for the lag coefficients y_{t-p}) and V (for the monthly dummy parameters d_t) and the structural impact matrix $\tilde{B}$ as follows:

$$MSE_{relative} = \exp \left(\frac{1}{n} \sum_{i=1}^n \ln(r_i) \right) \text{ where } r_i = \frac{TSE_i}{TSE_i^{p_0}} \quad (1)$$

We estimate these parameters using the original data for five different lag orders, $p_0 \in 4, 6, 8, 10$ and create DGPs with different sample sizes, $T \in 240, 300, 360, 480, 600$, where $T = N + \bar{p}$. $\bar{p}$ is the set of lag orders that are considered for both the IC and the BVAR estimations; it is uniformly set to $\bar{p} = 12$ to ensure that the true lag order p_0 is part of the set. These specifications directly follow Ivanov and Kilian (2005). We do not increase the number of lags or higher sample sizes in order to keep the comparison intact and to keep computational capacities available for the switch to sign restriction decomposition and the addition of BVAR. For sampling, we use random draws of length p_0 from the original dataset as initial values, then apply the parametric bootstrap using the SVAR described above and generate sequences of lengths T .

On each of these samples, VARs fitted by lag orders selected by both the three ICs as well as different specifications of BVAR are estimated, a fixed number of admissible IRFs found, and the median target IRF determined. We only include AIC, BIC and HQC as information criteria and omit the Likelihood Ratio tests and the Lagrange Multiplier test for lag selection as these have become practically irrelevant in the last 20 years for macroeconomic time series analysis and were also shown to perform poorly in Ivanov and Kilian (2005). We estimate all BVARs using both $p = p_{max} = 12$ and $p = p_0$ as the lag order, and can therefore compare how BVAR performs in a correctly specified model compared to a general longer lag order, as it would be applied on real data where the true lag order is unknown. The selection of the overall shrinkage parameter λ_1 is the key criterion to determine how loose or how tight the prior is applied and to what extent further lag

orders are shrunk towards 0. We apply $\lambda_1 \in \{0.05, 0.2, 0.35, 0.5, 0.65, 0.8, 0.95\}$ to determine how optimal shrinkage depends on sample size T and true lag order p_0 . Additionally, we use the marginal data density (MDD) method to determine a λ_1 optimized on Out-of-Sample MSE for which we also provide the same set of admissible λ_1 -values. The limitation of the value-field is established for computational reasons; it captures a good range of typically used λ_1 -values, both tighter and looser specifications.

Finally, the squared error matrix of the selected median target IRF to the true DGP IRF is calculated, collapsed into a scalar total squared error by summation. Then we estimate the ratio of this squared error of a specific estimation approach to the model fitted by the true lag order p_0 . Therefore, the deviations of the true IRF are directly put into relation to how well a correctly specified model compares for the given sample. Finally, we estimate the MSE by calculating the geometric mean of these individual ratios. The geometric mean is preferred over the arithmetic mean as the ratios are bounded on the left side at 0, but unbounded on the right side. Outliers on the right side are hence overweighed in the arithmetic mean. The relative MSE for one method for one true lag order p_0 and one sample size T is given by the following formula:

$$MSE_{relative} = \exp \left(\frac{1}{n} \sum_{i=1}^n \ln(r_i) \right) \text{ where } r_i = \frac{TSE_i}{TSE_{p_0}} \quad (2)$$

While, for a single estimation it is possible to estimate millions of admissible IRFs, the Monte Carlo simulation approach doesn't allow for such a large number of draws. In establishing the median target IRF specific to the DGP, we set the number of discovered admissible IRFs to 1 million so as to have a very accurate median target IRF for the specific dataset. In the Monte Carlo Simulation, we reduce this number drastically to 100 as it has proven to be the absolute limit that still allows us to have an acceptable number of Monte Carlo iterations, 1000, for our set of different methods. Even with computational optimization by parallel processing, more draws are not applicable under our hardware constraints. For each individual iteration, the median target IRF is therefore less accurately determined. However, because we use relative MSEs as our performance measure and repeat the process 1000 times, this should not impact the interpretability of our results.

3 Results

Present the results of the Monte Carlo simulations, broken down by the specific strategies employed to handle model uncertainty.

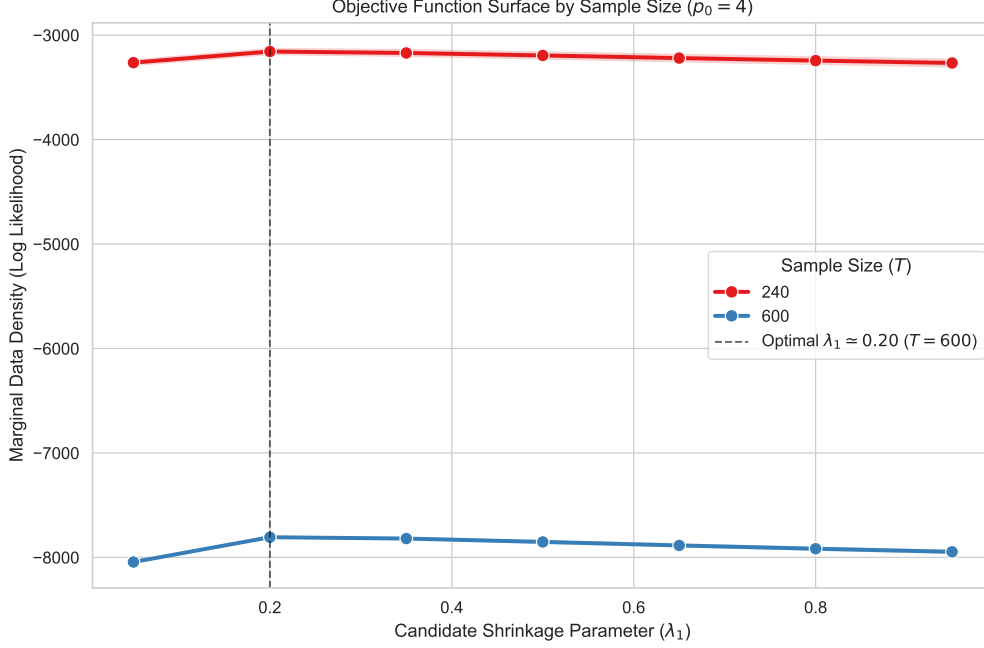


Figure 1: Objective Function Surface for Marginal Data Density Optimization. The peak represents the optimal balance between model fit and parameter penalty.

3.1 Behavior of Standard Information Criteria

As established theoretically as well as in the foundational Monte Carlo simulation study on information criterias, Lütkepohl (1985), AIC tends to select higher lag orders than HQC and BIC, and BIC typically has the strongest reduction in included lags. Our results confirm this as seen in 1.

3.2 The Shrinkage Strategy: Priors vs. Data

This section compares hard lag selection against dynamic BVARs. We analyze both the bias-variance tradeoff and how the optimal prior relaxes asymptotically as sample sizes increase.

2 illustrates how BVAR performs in comparison to BIC for $T = 240$ and $T = 600$ at $p_0 = 4$ at the different values of λ_1 .

Finally, we can compare how BVAR using the true lag order p_0 performs in comparison to BVAR with p_{max} . We include this to find out whether the shrinkage of BVAR is strong enough to reduce an overfitted model to achieve good results or whether a more accurate lag order selection is highly beneficial to the BVAR approach. Additionally, we can reveal whether optimal λ_1 -values change when the pre-determined lag order changes. 3 shows the relative MSE of the BVAR fitted with p_0 with the BVAR fitted with p_{max} . Cases where p_0 outperforms our baseline, i.e. $MSE < 1$ are shown in bold.

Table 1: Average Evaluated Lag Order ($\hat{p}$) selected by Information Criteria.

	Estimator	AIC	BIC	HQC
p_0	T			
4	240	3.00	2.00	2.50
	300	3.50	2.00	2.00
	360	2.50	2.00	2.00
	480	4.00	2.00	3.00
	600	3.50	2.00	3.00
6	240	5.50	2.00	2.00
	300	5.00	2.00	2.00
	360	6.50	2.00	2.00
	480	5.50	2.00	3.00
	600	6.00	2.00	2.50
8	240	8.00	2.00	2.00
	300	8.00	2.00	2.00
	360	8.00	2.00	2.00
	480	8.00	2.00	2.00
	600	8.00	2.00	3.50
10	240	2.50	2.00	2.00
	300	10.00	2.00	2.50
	360	8.50	2.00	2.00
	480	9.00	2.00	2.00
	600	9.50	2.50	2.50

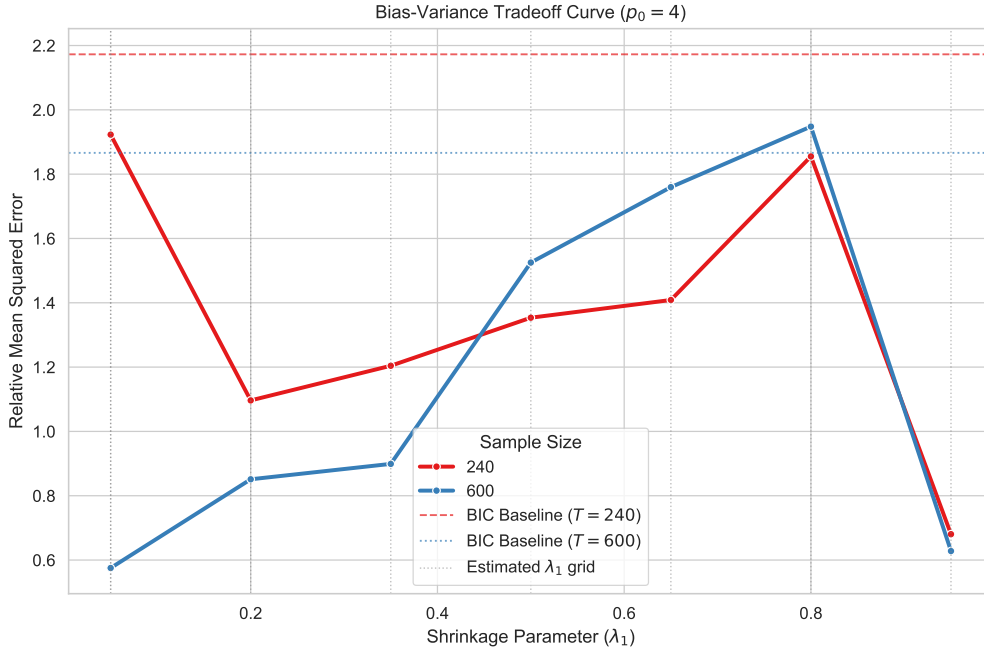


Figure 2: Comparison of Bias-Variance Tradeoff: BVAR vs. Hard Selection. The curve illustrates how the optimal τ balances bias and variance, achieving superior MSE performance compared to fixed lag selection.

Table 2: Relative Mean Squared Error: Hard Selection vs. Bayesian Shrinkage.

T	p_0	AIC	BIC	HQC	BVAR ($\lambda_1 = 0.05$)	BVAR ($\lambda_1 = 0.20$)	BVAR ($\lambda_1 = 0.35$)	BVAR ($\lambda_1 = 0.50$)	BVAR ($\lambda_1 = 0.65$)	BVAR ($\lambda_1 = 0.80$)	BVAR ($\lambda_1 = 0.95$)	BVAR (MDD λ_1)
240	4	1.671	2.173	1.801	1.575	1.416	0.728	0.772	0.740	1.093	0.879	1.416
	6	0.917	0.603	0.603	1.555	0.755	0.618	0.759	0.657	0.699	1.051	0.755
	8	1.000	0.399	0.399	0.698	0.785	0.459	0.600	0.872	1.012	0.947	0.785
	10	0.542	1.024	1.024	1.179	0.591	0.469	1.151	0.156	0.452	0.840	0.591
300	4	2.224	5.344	5.344	4.460	1.246	6.096	3.025	3.184	1.939	3.926	1.246
	6	0.368	1.212	1.212	0.515	1.082	1.013	0.848	1.033	2.770	0.797	1.082
	8	1.000	0.221	0.221	0.175	0.555	0.999	0.558	0.998	0.837	0.459	0.555
	10	1.503	5.044	1.803	2.777	1.727	3.610	1.501	3.534	3.319	1.783	3.372
360	4	3.881	1.419	1.419	3.272	0.459	1.561	1.845	1.576	0.951	2.407	0.459
	6	0.726	3.349	3.349	2.342	0.283	3.670	0.604	1.535	1.735	2.270	0.283
	8	1.000	2.905	2.905	2.949	3.511	1.417	2.170	2.385	2.320	2.628	3.511
	10	0.656	2.304	2.304	1.782	0.669	0.336	1.449	0.960	0.967	2.512	0.669
480	4	1.000	0.269	0.383	0.350	1.477	0.503	1.184	0.420	0.269	1.326	1.477
	6	0.935	2.243	2.239	4.763	2.229	2.062	1.836	2.191	2.175	1.740	2.229
	8	1.000	1.610	1.610	1.183	1.502	0.882	1.636	1.079	2.805	0.895	1.502
	10	2.747	1.737	1.737	8.680	1.707	2.136	1.721	1.402	3.574	5.206	1.598
600	4	1.203	1.866	1.220	2.119	1.271	1.744	1.532	2.391	1.529	0.772	1.271
	6	1.000	1.349	1.012	0.977	1.444	1.890	1.017	1.288	1.515	0.940	1.444
	8	1.000	0.709	0.543	0.354	0.362	1.419	0.344	1.203	1.153	0.330	0.786
	10	0.677	0.756	0.756	0.919	1.259	2.377	1.507	2.310	0.524	0.673	2.377

Table 3: Ratio of Relative MSE: Estimating with True Lag Order (p_0) vs. Maximum Lag Order ($p_{\max}$). Values < 1.0 indicate p_0 outperforms $p_{\max}$.

T	p_0	λ_1 0.05	λ_1 0.20	λ_1 0.35	λ_1 0.50	λ_1 0.65	λ_1 0.80	λ_1 0.95	MDD λ_1
240	4	0.126	0.583	1.508	1.592	1.961	0.749	1.035	0.583
	6	0.951	1.108	0.967	0.222	0.609	0.643	0.420	1.108
	8	2.460	2.385	3.265	3.089	1.030	0.802	1.676	2.385
	10	0.521	0.742	0.976	0.819	3.949	1.778	0.829	0.742
300	4	1.382	6.007	0.555	0.863	1.605	1.860	2.673	6.007
	6	3.113	1.072	2.131	0.977	1.369	0.595	1.077	1.072
	8	6.684	0.191	0.519	0.514	0.228	0.194	0.576	0.191
	10	1.071	0.919	0.440	1.749	0.642	0.520	0.668	0.472
360	4	0.674	2.783	0.086	0.061	0.641	1.471	0.318	2.783
	6	1.560	5.719	1.035	2.212	0.694	1.090	0.199	5.719
	8	0.380	0.100	2.063	0.449	0.531	0.453	0.139	0.100
	10	2.072	2.409	5.660	1.497	0.317	0.812	0.816	2.409
480	4	3.874	0.976	1.812	0.487	1.569	2.270	0.868	0.778
	6	0.152	1.073	2.227	0.736	1.306	1.038	2.858	2.466
	8	2.138	0.698	0.972	0.435	0.573	0.549	1.032	0.571
	10	0.334	2.382	1.723	1.308	1.285	1.214	0.838	2.866
600	4	1.007	0.631	0.310	0.537	1.138	0.673	1.032	0.179
	6	1.062	0.843	0.843	0.928	0.896	1.518	0.507	0.843
	8	0.494	1.419	1.198	0.640	0.935	0.624	2.137	1.075
	10	1.384	0.524	0.365	0.332	0.087	1.515	1.166	0.365

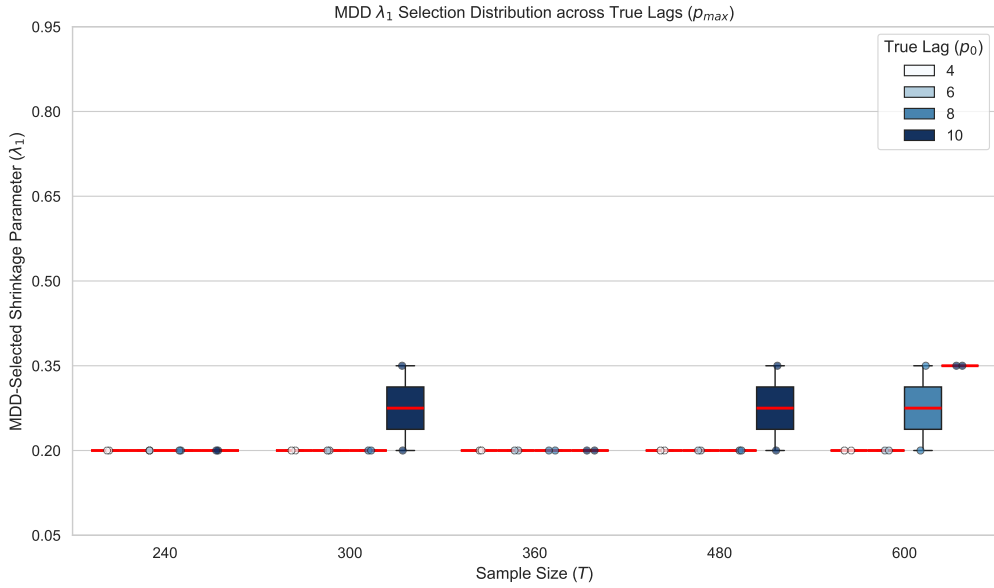


Figure 3: The MDD selected priors for different sample sizes for $p = p_{\max} = 12$, $p = p_0 = 4$.



Figure 4: The MDD selected priors for different sample sizes for $p = p_0 = 12$.

4 Conclusion

Summarize the main findings, specifically highlighting how optimal τ selection and true BMA systematically outperform standard econometric heuristics in finite samples.

References

References

- Gertler, Mark and Peter Karadi (Jan. 1, 2015). “Monetary Policy Surprises, Credit Costs, and Economic Activity”. In: *American Economic Journal: Macroeconomics* 7.1, pp. 44–76. ISSN: 1945-7707, 1945-7715. DOI: [10.1257/mac.20130329](https://pubs.aeaweb.org/doi/10.1257/mac.20130329). URL: <https://pubs.aeaweb.org/doi/10.1257/mac.20130329> (visited on 04/25/2026).
- Ivanov, Ventzislav and Lutz Kilian (Jan. 14, 2005). “A Practitioner’s Guide to Lag Order Selection For VAR Impulse Response Analysis”. In: *Studies in Nonlinear Dynamics & Econometrics* 9.1. ISSN: 1558-3708. DOI: [10.2202/1558-3708.1219](https://www.degruyter.com/document/doi/10.2202/1558-3708.1219/html). URL: <https://www.degruyter.com/document/doi/10.2202/1558-3708.1219/html> (visited on 04/22/2026).
- Kilian, Lutz (June 2009). “Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market”. In: *American Economic Review* 99.3, pp. 1053–1069. ISSN: 0002-8282. DOI: [10.1257/aer.99.3.1053](https://www.aeaweb.org/articles?id=10.1257/aer.99.3.1053). URL: <https://www.aeaweb.org/articles?id=10.1257/aer.99.3.1053> (visited on 06/19/2026).
- Kilian, Lutz and Daniel P. Murphy (Apr. 1, 2014). “The Role of Inventories and Speculative Trading in the Global Market for Crude Oil”. In: *Journal of Applied Econometrics* 29.3, pp. 454–478. ISSN: 1099-1255. DOI: [10.1002/jae.2322](https://onlinelibrary.wiley.com/doi/10.1002/jae.2322). URL: <https://onlinelibrary.wiley.com/doi/10.1002/jae.2322> (visited on 05/12/2026).
- Lütkepohl, Helmut (1985). “Comparison of Criteria for Estimating the Order of a Vector Autoregressive Process”. In: *Journal of Time Series Analysis* 6.1, pp. 35–52. ISSN: 1467-9892. DOI: [10.1111/j.1467-9892.1985.tb00396.x](https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9892.1985.tb00396.x). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9892.1985.tb00396.x> (visited on 04/22/2026).
- Uhlig, Harald (Mar. 2005). “What Are the Effects of Monetary Policy on Output? Results from an Agnostic Identification Procedure”. In: *Journal of Monetary Economics* 52.2, pp. 381–419. ISSN: 03043932. DOI: [10.1016/j.jmoneco.2004.05.007](https://linkinghub.elsevier.com/retrieve/pii/S0304393205000073). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0304393205000073> (visited on 06/11/2026).

Declaration of AI Usage

During the preparation of this seminar paper, I, [NAME], used Claude, Gemini, and ChatGPT for help with coding, research, and adapting writing to academic standards. I have reviewed and edited the content as needed and take full responsibility for the content.

[NAME]